

Rebuilding Rosei's Interband ϵ_2 of Gold in His Own Notation

From Fermi's Golden Rule in k-space — recreating Guerrisi–Rosei–Winsemius Eqs. (1)–(10),
and fixing the three printed ones that are wrong: (4), (6), (8)

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Before the algebra: what this note is, and where it goes

This note rebuilds Rosei's interband $\epsilon_2(\hbar\omega, T)$ of gold from the ground up — starting at Fermi's Golden Rule, in Rosei's own symbols — so that every one of the ten numbered equations in Guerrisi, Rosei & Winsemius (“GRW”, *Phys. Rev. B* **12**, 557 (1975)) drops out of a single, continuous derivation. I wrote it this way because I wanted one place where the whole chain is visible, and because along the way three of GRW's printed equations turn out to be wrong; I would rather fix them out in the open than quietly use corrected forms.

Four things to keep in your head before any algebra:

1. Where we are heading. GRW's two headline results are their Eq. (7) and their Eq. (9). Everything else in the paper (and in this note) exists only to feed these two. Eq. (7) is the thermally weighted, energy-integrated joint density of states,

$$\mathcal{J}_{l \rightarrow u}(\hbar\omega, T) = \int_{E_{\min}}^{E_{\max}} D_{l \rightarrow u}(E, \hbar\omega) [1 - f(E, T)] dE, \quad (\text{GRW 7})$$

and Eq. (9) turns that into the measurable quantity,

$$\epsilon_2(\hbar\omega, T) = \frac{8\pi^2 e^2 \hbar^4}{3m^2 (\hbar\omega)^2} \left[|P_X(l \rightarrow u)|^2 g_X(\hbar\omega, T) + |P_L(l \rightarrow u)|^2 g_L(\hbar\omega, T) \right], \quad g_i = N_i \mathcal{J}_i. \quad (\text{GRW 9})$$

So the plan is: build $D_{l \rightarrow u}$ (the density of transitions), integrate it against occupation to get $\mathcal{J}$ (Eq. 7), and slot that into ϵ_2 (Eq. 9). If you only remember two equations from Rosei, remember these.

2. What Rosei actually derives here, and what he just borrows. This matters, and it is easy to miss. *In this 1975 paper Rosei genuinely works out only the X point* — the saddle, and its unusual sub-gap tail; that is the new physics of the paper. *The L point he does not derive at all*: in his fitting section he writes, in so many words, that “the procedure for obtaining the JDOS for transitions at L has been presented in a number of papers and will not be repeated here,” pointing back to his own earlier work (his Refs. 10 and 12, i.e. Rosei 1974). Even the *general recipe* for the density of transitions — his Eq. (4) — he introduces with “following the procedure outlined in Ref. 10.” So the split is: X is the fresh work of this paper; L and the EDJDOS machinery are inherited. I re-derive L here anyway, with the same tools, so that both absorption edges live in one self-contained framework rather than scattered across three papers.

3. Three of GRW's printed equations don't survive the rederivation, and I fix them — most likely they are just typos in the 1975 print. Two — Eqs. (6) and (8) — are

plainly transcription slips: (6) is dimensionally impossible as printed, and (8) has two subscripts interchanged. The third, Eq. (4), is subtler: its prefactor comes out off by a constant factor for *any* sensible matrix-element convention. Each is flagged in a **red callout box** at the point it comes up, with the printed form shown next to the correction so you can see exactly what changed.

4. Conventions. Gaussian units throughout (as in the 1975 paper); E is measured from the Fermi level, $E_F = 0$. And one point that trips people up, spelled out in Sec. 1 below: *every energy level and every effective mass in this note is a positive number* — a magnitude — with all the signs written out explicitly in front of the terms.

Contents

1	Notation, and a word on signs	2
2	Fermi’s Golden Rule in k-space	3
3	The X point — the part Rosei actually derives	3
3.1	Bands and the constant-energy-difference surface: GRW (1)–(3)	3
3.2	The squared-momentum substitution and the Jacobian	4
3.3	The energy-distributed JDOS: GRW (4)–(5), with (4) corrected	5
3.4	Topology, the integration limits, and GRW (7)–(8)	6
4	The L point — the part Rosei borrows	8
5	Putting it together: GRW (9)–(10)	9
6	Photonic density of states and the emission spectrum	10
6.1	Mode counting	10
6.2	Spontaneous emission of one pair	10
6.3	The interband PL spectrum, and the closure check	11
A	Dimensional audit (Gaussian)	11

1 Notation, and a word on signs

E	final-state (upper-band) energy, from $E_F = 0$; $\hbar\omega$: photon energy
$\hbar\omega_{X_6^-}, \hbar\omega_{X_7^+}$	level distances from E_F at X (X_6^- above, X_7^+ below); likewise $\hbar\omega_{L_6^-}, \hbar\omega_{L_+}$
$\mathcal{E}_g^X, \mathcal{E}_g^L$	direct gaps, $\mathcal{E}_g^X = \hbar\omega_{X_6^-} + \hbar\omega_{X_7^+}$, $\mathcal{E}_g^L = \hbar\omega_{L_6^-} + \hbar\omega_{L_+}$
$m_{u\perp}, m_{u\parallel}, m_{l\perp}, m_{l\parallel}$	effective masses (positive; signs written explicitly)
A_i, B_i	$\hbar^2/2m_{i\perp}, \hbar^2/2m_{i\parallel}$ ($i = u, l$): curvature shorthands for derivations
$\Omega_{lu}(\mathbf{k})$	transition energy $\hbar\omega_u - \hbar\omega_l$
$D_{l\rightarrow u}, \mathcal{F}_{l\rightarrow u}, \mathcal{D}_{l\rightarrow u}$	EDJDOS, its mass parameter, and the Jacobian determinant
$\mathcal{J}(\hbar\omega, T)$	thermally weighted EDJDOS integral, GRW Eq. (7)
$ P ^2$	$ \langle u \hat{\mathbf{e}}\cdot\nabla l\rangle ^2$ averaged (length $^{-2}$; Sec. 2)
$N_X = 6, N_L = 8$	equivalent half-neighborhoods (GRW’s counting)
$f(E, T), n_B(\hbar\omega)$	Fermi–Dirac and Bose–Einstein functions

Gaussian units (e^2 has dimension energy $\times$ length), matching the 1975 paper.

Are the level energies and masses “absolute values”? Yes — and this is worth pinning down before we write a single dispersion relation. Every level energy here — $\hbar\omega_{X_6^-}, \hbar\omega_{X_7^+}, \hbar\omega_{L_6^-}, \hbar\omega_{L_+}$ — is a *positive number*, a distance from the Fermi level. Every effective mass — $m_{u\perp}, m_{u\parallel}, m_{l\perp}, m_{l\parallel}$ — is likewise a *positive number*, a magnitude. There are no signed masses and no signed levels anywhere in this note. All the physics of *which way a band curves*

and *on which side of E_F a level sits* is carried by the explicit $\pm$ written in front of each term, never hidden inside the symbol. So in Eq. (3.1) below, the leading $+\hbar\omega_{X_6^-}$ says the upper level is *above* E_F , and the $-\hbar^2 k_{\parallel}^2/2m_{u\parallel}$ term says the band *falls* along Δ — with $m_{u\parallel}$ itself a plain positive number.

One place this can cause confusion: the Christensen & Seraphin (C&S) mass table I extract the curvatures from lists some masses *with* a minus sign — e.g. the X sp -band longitudinal curvature as $m_{u\parallel} \simeq -0.40$. That minus is just C&S’s shorthand for “this band curves downward here.” In the present notation the same curvature is a positive $m_{u\parallel} = 0.40$ with the minus pulled out into the explicit operator sign of Eq. (3.1). Same physics, sign bookkeeping moved from the symbol to the term.

2 Fermi’s Golden Rule in $\mathbf{k}$ -space

We start where any optical-absorption calculation in a solid starts. Minimal coupling in the Coulomb gauge gives the light–matter interaction $\hat{H}_{\text{int}} = (e/mc) \mathbf{A} \cdot \hat{\mathbf{p}}$ (the A^2 term drives no inter-band transition and is dropped). For a classical wave $\mathbf{E} = \hat{\mathbf{e}} E_0 \cos \omega t$, i.e. $\mathbf{A} = -(cE_0/\omega) \hat{\mathbf{e}} \sin \omega t$, the absorption half of $\hat{H}_{\text{int}}$ has amplitude $(eE_0/2m\omega) \hat{\mathbf{e}} \cdot \hat{\mathbf{p}}$. Photon momentum is negligible on the scale of the zone, so transitions are *vertical* (same $\mathbf{k}$ up and down). Apply FGR per initial Bloch state, sum over $\mathbf{k}$ (keeping spin explicit), and set the result equal to the average dissipated power $\langle \dot{U} \rangle = \omega \epsilon_2 E_0^2 / 8\pi$ per unit volume. That balance gives the exact starting point,

$$\epsilon_2(\omega) = \frac{4\pi^2 e^2}{m^2 \omega^2} \frac{2}{(2\pi)^3} \int_{\text{BZ}} d^3 k |\hat{\mathbf{e}} \cdot \mathbf{p}_{ul}(\mathbf{k})|^2 \delta(\hbar\omega_u(\mathbf{k}) - \hbar\omega_l(\mathbf{k}) - \hbar\omega) [f(\hbar\omega_l) - f(\hbar\omega_u)]. \quad (2.1)$$

Near a critical point Rosei freezes the matrix element (it barely varies over the tiny active region) and averages over polarization. Writing the momentum matrix element in the older gradient form, $\mathbf{p}_{ul} = -i\hbar \langle u | \nabla | l \rangle$, and defining

$$|P|^2 \equiv |\langle u | \nabla | l \rangle|^2 \quad (\text{dimension length}^{-2}), \quad \overline{|\hat{\mathbf{e}} \cdot \mathbf{p}_{ul}|^2} = \frac{\hbar^2 |P|^2}{3}, \quad (2.2)$$

Eq. (2.1) becomes, per critical-point family,

$$\epsilon_2(\hbar\omega, T) = \frac{4\pi^2 e^2 \hbar^2}{3m^2 \omega^2} \cdot \frac{2}{(2\pi)^3} N |P|^2 \int_{\text{half-nbhd}} d^3 k \delta(\Omega_{lu}(\mathbf{k}) - \hbar\omega) [f(E - \hbar\omega) - f(E)] \Big|_{E=\hbar\omega_u(\mathbf{k})}, \quad (2.3)$$

where N counts equivalent *half*-neighborhoods ($N_X = 6$, $N_L = 8$: each zone face carries half of a full constant-energy- difference surface). Keep this half-counting paired with the half-space density of states below, or you silently pick up a factor of 2. The occupation factor is the full $f(E - \hbar\omega) - f(E)$; for d bands sitting $\gtrsim 1.5$ eV below E_F it collapses to GRW’s $[1 - f(E, T)]$, but I keep the general form so the same expression survives into the non-equilibrium extensions (f^S for CW, f^P for pulsed).

3 The X point — the part Rosei actually derives

This is the heart of the 1975 paper, so it gets the full treatment. The X point is a *saddle*, and the payoff of doing it carefully is the sub-gap tail: absorption that starts *below* the nominal gap, purely from geometry, with no broadening invoked.

3.1 Bands and the constant-energy-difference surface: GRW (1)–(3)

GRW’s parabolic band pair at X , exactly as they write it (energies from E_F ; $k_{\parallel}$ runs along $\Delta = \Gamma X$, $k_{\perp}$ lies in the zone face). The upper X_6^- (sp) band curves *up* in the face and *down* along

Δ — a saddle:

$$E = \hbar\omega_u(\mathbf{k}) = \hbar\omega_{X_6^-} + \frac{\hbar^2 k_\perp^2}{2m_{u\perp}} - \frac{\hbar^2 k_\parallel^2}{2m_{u\parallel}}, \quad [\text{GRW (1)}] \quad (3.1)$$

$$E - \hbar\omega = \hbar\omega_l(\mathbf{k}) = -\hbar\omega_{X_7^+} - \frac{\hbar^2 k_\perp^2}{2m_{l\perp}} - \frac{\hbar^2 k_\parallel^2}{2m_{l\parallel}}, \quad [\text{GRW (2)}] \quad (3.2)$$

and the lower X_7^+ (d) band is a plain maximum (curves down both ways). (Reading off the signs, exactly as promised in Sec. 1: $\hbar\omega_{X_6^-}$ enters +, so X_6^- is above E_F ; $\hbar\omega_{X_7^+}$ enters −, so X_7^+ is below; the one minus in front of the $k_\parallel^2$ term of (3.1) is what makes the upper band a saddle rather than a bowl.) Vertical transitions of energy $\hbar\omega$ live on the constant-energy-difference surface (CEDS),

$$\Omega_{lu}(\mathbf{k}) - \hbar\omega = \hbar\omega_u(\mathbf{k}) - \hbar\omega_l(\mathbf{k}) - \hbar\omega = 0. \quad [\text{GRW (3)}] \quad (3.3)$$

Subtract (3.2) from (3.1), with the curvature shorthands $A_i = \hbar^2/2m_{i\perp}$, $B_i = \hbar^2/2m_{i\parallel}$:

$$\Omega_{lu} = \mathcal{E}_g^X + \bar{A}_X k_\perp^2 + \bar{B}_X k_\parallel^2, \quad \bar{A}_X = A_u + A_l > 0, \quad \bar{B}_X = B_l - B_u. \quad (3.4)$$

The sign of $\bar{B}_X$ decides the whole X story. GRW’s own apparatus — the open CEDS, the sub-gap tail, the $-20k_B T$ floor of their Eq. (7), the step-like onset near 1.8 eV — only makes sense if $\boxed{\bar{B}_X < 0}$, i.e. $m_{u\parallel} < m_{l\parallel}$: the sp band falls along Δ *faster* than the d band (the “flat- d ” picture). The main text follows this branch; my own C&S extraction currently disagrees on this one assignment, which I flag honestly in Remark 3.1 rather than bury.

3.2 The squared-momentum substitution and the Jacobian

Here is the one trick the whole reduction rests on. Set

$$u = k_\perp^2, \quad v = k_\parallel^2 \quad (u, v \geq 0).$$

I used to call this a “linearization,” but that name is misleading and I’ve dropped it: nothing is being approximated. It is an *exact* change of variable. In these squared-momentum variables the two conditions that pin a transition down become honest *straight-line* relations — and, importantly, **each row of the matrix below is simply one of the equations we already have, with the coefficients of $(u, v) = (k_\perp^2, k_\parallel^2)$ read straight off:**

- the *top* row is the final-state energy, i.e. Eq. (1), (3.1): $E - \hbar\omega_{X_6^-} = A_u k_\perp^2 - B_u k_\parallel^2 = A_u u - B_u v$;
- the *bottom* row is the transition energy $\hbar\omega$, i.e. the CEDS condition Eq. (3), (3.3), written out with (3.4): $\hbar\omega - \mathcal{E}_g^X = \bar{A}_X k_\perp^2 + \bar{B}_X k_\parallel^2 = \bar{A}_X u + \bar{B}_X v$.

Stacking those two equations *is* the construction of the transition matrix M , and the Jacobian of the map is nothing but its determinant:

$$\begin{pmatrix} E - \hbar\omega_{X_6^-} \\ \hbar\omega - \mathcal{E}_g^X \end{pmatrix} = \underbrace{\begin{pmatrix} A_u & -B_u \\ \bar{A}_X & \bar{B}_X \end{pmatrix}}_M \begin{pmatrix} u \\ v \end{pmatrix}, \quad \det M = A_u \bar{B}_X + B_u \bar{A}_X = A_u B_l + A_l B_u \equiv \mathcal{D}_X > 0. \quad (3.5)$$

So $\mathcal{D}_X \equiv \det M$ is exactly the Jacobian that will collapse the two δ -functions in (3.8) below; it is built entirely from (3.1) and (3.3). Note it is positive for *either* sign of $\bar{B}_X$ — a small but useful fact. Inverting the 2×2 system,

$$u = \frac{\bar{B}_X (E - \hbar\omega_{X_6^-}) + B_u (\hbar\omega - \mathcal{E}_g^X)}{\mathcal{D}_X}, \quad v = k_\parallel^2 = \frac{A_u (\hbar\omega - \mathcal{E}_g^X) - \bar{A}_X (E - \hbar\omega_{X_6^-})}{\mathcal{D}_X}. \quad (3.6)$$

The second line is the whole content of GRW's Eq. (6). Restoring it to GRW's mass symbols via $\mathcal{D}_X = (\hbar^2/2)^2 \mathcal{F}^{-2}$ (Eq. (3.11) below),

$$k_{\parallel}^2 = \frac{2\mathcal{F}^2}{\hbar^2} \left[\frac{\hbar\omega - \hbar\omega_{X_7^+} - E}{m_{u\perp}} + \frac{\hbar\omega_{X_6^-} - E}{m_{l\perp}} \right]. \quad [\text{GRW (6) corrected}] \quad (3.7)$$

The printed Eq. (6) can't stand as written — almost certainly a typesetting slip.

GRW print this line as

$$k_{\parallel} = \left(\hbar\omega - \hbar\omega_{X_7^+} + \frac{\hbar^2}{2m_{l\perp}} (\hbar\omega_{X_6^-} - E) - \frac{\hbar^2}{2m_{u\perp}} E \right)^{1/2}. \quad (6, \text{ as printed})$$

You can see the trouble without doing any physics: it sets a wavevector $k_{\parallel}$ equal to (the square root of) a sum of *energies*. Inside the bracket it also *adds* a bare energy $\hbar\omega - \hbar\omega_{X_7^+}$ to terms of the form $(\hbar^2/2m) \times \text{energy}$, which don't even share its units. The coefficients simply read $\hbar^2/2m$ where they should read $2m/\hbar^2$ for the bracket to come out as $k_{\parallel}^2 \sim \text{length}^{-2}$. Three small fixes turn the printed line into (3.7): (i) square the left-hand side ($k_{\parallel} \rightarrow k_{\parallel}^2$); (ii) invert the coefficients and pull out the overall $2\mathcal{F}^2/\hbar^2$; (iii) restore the missing $1/m_{u\perp}$ on the first term. The three numerator slots then line up with the print term by term, which is how I'm confident (3.7) is what was meant — most likely just a typo, but a disabling one, since the printed formula is unusable as written.

3.3 The energy-distributed JDOS: GRW (4)–(5), with (4) corrected

Rosei's central object is the number of $l \rightarrow u$ transition pairs, per unit crystal volume, per unit final energy E , per unit transition energy $\hbar\omega$, for one half-neighborhood and one spin:

$$D_{l \rightarrow u}(E, \hbar\omega) \equiv \frac{1}{(2\pi)^3} \int_{k_{\parallel} > 0} d^3k \, \delta(\hbar\omega_u(\mathbf{k}) - E) \delta(\Omega_{lu}(\mathbf{k}) - \hbar\omega). \quad (3.8)$$

Let me do this reduction all the way, because it is where both $\mathcal{D}_X$ and Rosei's $\mathcal{F}$ actually come from. Axial symmetry about ΓX gives $d^3k = 2\pi k_{\perp} dk_{\perp} dk_{\parallel}$; in the squared variables $k_{\perp} dk_{\perp} = \frac{1}{2} du$ and $dk_{\parallel} = dv/(2\sqrt{v})$, so the measure becomes

$$\frac{1}{(2\pi)^3} \int_{k_{\parallel} > 0} d^3k = \frac{1}{(2\pi)^3} \frac{\pi}{2} \int_0^\infty du \int_0^\infty \frac{dv}{\sqrt{v}} = \frac{1}{16\pi^2} \int_0^\infty du \int_0^\infty \frac{dv}{\sqrt{v}}.$$

The two δ -functions in (3.8) are *exactly the two rows of the transition matrix* (3.5): their arguments are $A_u u - B_u v - (E - \hbar\omega_{X_6^-})$ (top row) and $\bar{A}_X u + \bar{B}_X v - (\hbar\omega - \mathcal{E}_g^X)$ (bottom row). A pair of δ 's whose arguments are linear in (u, v) collapses onto the unique solution (u_*, v_*) of that system — i.e. onto (3.6) — dividing by the modulus of the Jacobian, which is precisely $|\det M| = \mathcal{D}_X$:

$$\delta(A_u u - B_u v - (E - \hbar\omega_{X_6^-})) \delta(\bar{A}_X u + \bar{B}_X v - (\hbar\omega - \mathcal{E}_g^X)) = \frac{1}{\mathcal{D}_X} \delta(u - u_*) \delta(v - v_*).$$

Doing the u, v integrals sets $\sqrt{v_*} = k_{\parallel}(E, \hbar\omega)$ from (3.7), and everything telescopes to

$$D_{l \rightarrow u}(E, \hbar\omega) = \frac{1}{16\pi^2 \mathcal{D}_X} \frac{1}{k_{\parallel}(E, \hbar\omega)}. \quad (3.9)$$

This line is the entire result of the reduction, and notice that it contains only the Jacobian determinant $\mathcal{D}_X$ and $k_{\parallel}$ — there is no $\mathcal{F}$ in it yet. Rosei's $\mathcal{F}_{l \rightarrow u}$ is *not* an

independent quantity that I am free to insert; it is nothing but a repackaging of $\mathcal{D}_X$, and here is where it is manufactured. Expand $\mathcal{D}_X$ in the masses using $A_i = \hbar^2/2m_{i\perp}$, $B_i = \hbar^2/2m_{i\parallel}$:

$$\mathcal{D}_X = A_u B_l + A_l B_u = \left(\frac{\hbar^2}{2}\right)^2 \left[\frac{1}{m_{u\perp} m_{l\parallel}} + \frac{1}{m_{l\perp} m_{u\parallel}} \right] = \left(\frac{\hbar^2}{2}\right)^2 \frac{m_{l\perp} m_{u\parallel} + m_{l\parallel} m_{u\perp}}{m_{l\perp} m_{l\parallel} m_{u\perp} m_{u\parallel}}. \quad (3.10)$$

Now *define* Rosei's mass parameter as the pure shorthand $\mathcal{F}_{l\rightarrow u} \equiv \hbar^2/(2\sqrt{\mathcal{D}_X})$. Substituting (3.10), the factor $\hbar^2/2$ cancels top and bottom and the explicit mass form falls right out — this *is* GRW's Eq. (5), now derived rather than assumed:

$$\mathcal{F}_{l\rightarrow u} \equiv \frac{\hbar^2}{2\sqrt{\mathcal{D}_X}} = \left[\frac{m_{l\perp} m_{u\parallel} + m_{l\parallel} m_{u\perp}}{m_{l\perp} m_{l\parallel} m_{u\perp} m_{u\parallel}} \right]^{-1/2}. \quad [\text{GRW (5)}] \quad (3.11)$$

Reading the same definition the other way, $\mathcal{D}_X = \hbar^4/(4\mathcal{F}_{l\rightarrow u}^2)$, and substituting it back into (3.9) rewrites the EDJDOS in Rosei's own notation:

$$D_{l\rightarrow u}(E, \hbar\omega) = \frac{1}{16\pi^2 \mathcal{D}_X} \frac{1}{k_{\parallel}(E, \hbar\omega)} = \frac{\mathcal{F}_{l\rightarrow u}^2}{4\pi^2 \hbar^4} \frac{1}{k_{\parallel}(E, \hbar\omega)} \quad [\text{GRW (4) corrected}] \quad (3.12)$$

supported on $u, v \geq 0$, with $k_{\parallel}$ from (3.7). So to be completely clear: $\mathcal{F}$ is never used as an input. The Jacobian hands me $\mathcal{D}_X$; (3.11) is the one-line renaming of it; and $\mathcal{F}$ appears in (3.12) only because I chose to write $\mathcal{D}_X$ that way. Eq. (5) comes out untouched; Eq. (4) does not.

The printed Eq. (4): the prefactor comes out off by a constant — most likely a typo in the paper.

GRW print

$$D_{l\rightarrow u}(E, \hbar\omega) = (8\pi^2 \hbar^2)^{-1} \mathcal{F}_{l\rightarrow u} k_{\parallel}^{-1}. \quad (4, \text{ as printed})$$

This one is subtler than a dropped symbol. If $|P|^2$ is a squared gradient matrix element (or a squared momentum, the other common choice), then the pair {printed (4), printed (9)} doesn't balance dimensionally, whereas (3.12) — which is *forced*, with nothing left to choose, by the definition (3.8) — closes the FGR chain exactly all the way to ϵ_2 (Sec. 5). The two forms differ only by

$$\frac{\text{printed (4)}}{\text{correct (3.12)}} = \frac{2\mathcal{F}_{l\rightarrow u}}{\hbar^2},$$

a *constant for each critical point*. I can't tell from the paper alone whether that is a typesetting slip in the prefactor or just a convention I'm not matching — but it makes no difference to any result: a constant per critical point cannot bend a line shape, so it is absorbed into the fitted strength $S = \mathcal{F}|P|^2$ of GRW's Eq. (10). Every published number keeps its meaning; only the bookkeeping between D , $\mathcal{F}$ and $|P|^2$ shifts. What the derivation above settles is simply that (3.12) is the form consistent with the definition (3.8).

3.4 Topology, the integration limits, and GRW (7)–(8)

Now the part that makes X physically interesting. On GRW's branch $\bar{B}_X < 0$ the CEDS is an *open* hyperboloid for every $\hbar\omega$: transitions exist on *both* sides of $\mathcal{E}_g^X$, and along the CEDS the final energy E decreases without bound as you move away from the top edge ($dE \propto -\mathcal{D}_X dv < 0$). Because nothing bounds the window from below geometrically, it is cut from below by *occupation* alone — GRW use the practical thermal floor $E_{\min} = -20k_B T$, below which $[1 - f] \rightarrow 0$ kills the integrand. The upper edge $E_{\max}(\hbar\omega)$ is geometric, and it changes character at the gap:

Above the gap ($\hbar\omega \geq \mathcal{E}_g^X$): the edge is the $k_{\parallel} \rightarrow 0$ circle ($v = 0$ in (3.6)), where $D_{l \rightarrow u}$ of (3.12) diverges like an inverse square root:

$$E_{\max}^> = \hbar\omega_{X_6^-} + \frac{A_u}{\bar{A}_X} (\hbar\omega - \mathcal{E}_g^X) = \hbar\omega_{X_6^-} + \frac{m_{l\perp}}{m_{u\perp} + m_{l\perp}} (\hbar\omega - \mathcal{E}_g^X), \quad k_{\parallel}^2 = \frac{\bar{A}_X}{\mathcal{D}_X} (E_{\max}^> - E). \quad (3.13)$$

Below the gap ($\hbar\omega < \mathcal{E}_g^X$): now the $k_{\parallel} = 0$ circle simply does not exist ($v > 0$ everywhere on the CEDS); the edge is instead the $k_{\perp} = 0$ point ($u = 0$ in (3.6)), where $k_{\parallel}$ stays finite and $D_{l \rightarrow u}$ is finite:

$$E_{\max}^< = \hbar\omega_{X_6^-} - \frac{B_u}{|\bar{B}_X|} (\mathcal{E}_g^X - \hbar\omega) = \hbar\omega_{X_6^-} + (\mathcal{E}_g^X - \hbar\omega) \frac{m_{l\parallel}}{m_{u\parallel} - m_{l\parallel}}. \quad [\text{GRW (8) corrected}] \quad (3.14)$$

The printed Eq. (8) has two subscripts interchanged — again, most likely a typo.

GRW print the upper limit as

$$E_{\max} = \hbar\omega_{X_6^-} + (\hbar\omega_{X_7^+} + \hbar\omega_{X_6^-} - \hbar\omega) \frac{m_{u\parallel}}{m_{l\parallel} - m_{u\parallel}} = \hbar\omega_{X_6^-} + (\mathcal{E}_g^X - \hbar\omega) \frac{m_{u\parallel}}{m_{l\parallel} - m_{u\parallel}}. \quad (8, \text{ as printed})$$

The two parallel-mass subscripts are interchanged: the print carries $m_{u\parallel}/(m_{l\parallel} - m_{u\parallel})$ where the clean derivation gives $m_{l\parallel}/(m_{u\parallel} - m_{l\parallel})$ — one consistent $u \leftrightarrow l$ swap. (It's specific to (8): the subscripts of (6) are *not* swapped, so it isn't the same slip repeated.) The tell here is physical, not just algebraic. On the $\bar{B}_X < 0$ branch $m_{u\parallel} < m_{l\parallel}$, so the printed denominator $m_{l\parallel} - m_{u\parallel} > 0$ makes $E_{\max}$ climb *above* $\hbar\omega_{X_6^-}$ as $\hbar\omega$ drops below the gap — but the sub-gap CEDS has *no states* up there. The corrected (3.14) instead puts the sub-gap edge sensibly *below* $\hbar\omega_{X_6^-}$, which is what makes the tail work.

That finite sub-gap edge (3.14) is exactly why the X onset is *step-like* (GRW's own word) instead of a sharp divergence: at $T = 0$ absorption switches on when $E_{\max}^<$ crosses E_F ,

$$\hbar\omega_{\text{on}} = \mathcal{E}_g^X - \hbar\omega_{X_6^-} \frac{m_{l\parallel} - m_{u\parallel}}{m_{l\parallel}} \simeq 1.94 - 0.17 \times \frac{0.40 - 0.15}{0.40} = 1.83 \text{ eV}, \quad (3.15)$$

which is the piezomodulation threshold, with the famous low-energy tail filling 1.83–1.94 eV purely geometrically (again: no broadening put in by hand).

And finally the thermally weighted integral, verbatim as GRW write it — this is Eq. (7), the first of the two headline results from the opening:

$$\mathcal{J}(\hbar\omega, T) = \int_{E_{\min}}^{E_{\max}} D_{l \rightarrow u}(E, \hbar\omega) [1 - f(E, T)] dE, \quad [\text{GRW (7)}] \quad (3.16)$$

with $D_{l \rightarrow u}$ from (3.12), $E_{\max}$ from (3.13)/(3.14), $E_{\min} = -20k_B T$, and $[1 - f] \rightarrow f(E - \hbar\omega) - f(E)$ once we go out of equilibrium.

Remark 3.1 (The one mass assignment I'm not sure of at X) *Honesty check, because it changes a sign. My extraction of the CES Fig. 5 curvatures assigns $m_{u\parallel} = 0.40$, $m_{l\parallel} = 0.15$ — the opposite $\parallel$ ordering to GRW's branch — which gives $\bar{B}_X > 0$: a closed window $[\hbar\omega_{X_6^-} - \frac{B_u}{\bar{B}_X} (\hbar\omega - \mathcal{E}_g^X), E_{\max}^>]$, no geometric sub-gap tail, and a $\sqrt{\hbar\omega - \mathcal{E}_g^X}$ onset sitting exactly at $\mathcal{E}_g^X$. The catch is that above the gap both branches give the same line shape with the same singular edge (3.13) — the $\parallel$ masses enter only through $\mathcal{F}_X$, which the fit absorbs into S_X — so an optical fit alone cannot tell the two branches apart. Only the sub-gap physics and the extracted $|P_X|^2$ can. Both orderings of my extraction happen to make the d band the lighter one, which is against the flat- d prior, so I suspect a transcription swap*

in my own mass table is the likeliest fix. Until C&S Fig. 5 is re-measured, this note follows GRW ($\bar{B}_X < 0$) and flags every spot where the branch matters.

4 The L point — the part Rosei borrows

The single most important thing to say about L up front: Rosei does not derive it in the 1975 paper. In his fitting section he states plainly that the L procedure “has been presented in a number of papers and will not be repeated here,” citing his own earlier work (Refs. 10 and 12; the machinery is Rosei 1974, *Phys. Rev. B* **10**, 474). So while X was fresh work, L is inherited wholesale. I re-derive it here anyway — the same squared-momentum substitution gives it in three lines — so that both edges sit in one framework and you can see *exactly* where L differs from X . And it differs in essentially one sign.

At L the L_6^- conduction band is a *minimum* in both local directions ($k_{\parallel}$ along ΓL) — not a saddle — and the top d band is again a maximum:

$$E = \hbar\omega_u(\mathbf{k}) = \hbar\omega_{L_6^-} + \frac{\hbar^2 k_{\perp}^2}{2m_{u\perp}} + \frac{\hbar^2 k_{\parallel}^2}{2m_{u\parallel}}, \quad (4.1)$$

$$E - \hbar\omega = \hbar\omega_l(\mathbf{k}) = -\hbar\omega_{L^+} - \frac{\hbar^2 k_{\perp}^2}{2m_{l\perp}} - \frac{\hbar^2 k_{\parallel}^2}{2m_{l\parallel}}. \quad (4.2)$$

The only change from X is the sign of the upper-band $k_{\parallel}^2$ term: $+$ here ((4.1)), $-$ at X ((3.1)). That one flip cascades through everything:

$$\Omega_{lu} = \mathcal{E}_g^L + \bar{A}_L k_{\perp}^2 + \bar{B}_L k_{\parallel}^2, \quad \bar{B}_L = B_u + B_l > 0 \text{ unconditionally.} \quad (4.3)$$

Because $\bar{B}_L > 0$ *no matter what the masses do*, the CEDS is a *closed ellipsoid*, existing only for $\hbar\omega \geq \mathcal{E}_g^L$: **there is no sub-gap tail at L** — the sharp-versus-tailed contrast between the two edges is baked in right here. The straight-line system is (3.5) with $-B_u \rightarrow +B_u$ and $\bar{B}_X \rightarrow \bar{B}_L$, and the curvature determinant comes out as a *difference* rather than a sum (the ellipsoid’s fingerprint):

$$\mathcal{D}_L = A_u B_l - A_l B_u, \quad \mathcal{F}_L = \frac{\hbar^2}{2\sqrt{|\mathcal{D}_L|}} = \left[\frac{|m_{l\perp} m_{u\parallel} - m_{l\parallel} m_{u\perp}|}{m_{l\perp} m_{l\parallel} m_{u\perp} m_{u\parallel}} \right]^{-1/2}. \quad (4.4)$$

For gold the two products are $m_{l\perp} m_{u\parallel} = 0.70 \times 0.12 = 0.084$ and $m_{l\parallel} m_{u\perp} = 1.03 \times 0.24 = 0.247$, so $\boxed{\mathcal{D}_L < 0}$. Running the inversion (3.6) with the sign of $\mathcal{D}$ reversed (which flips both positivity inequalities) gives

$$k_{\parallel}^2 = \frac{\bar{A}_L}{|\mathcal{D}_L|} (E - E_{\min}), \quad D_{l \rightarrow u}(E, \hbar\omega) = \frac{1}{16\pi^2 |\mathcal{D}_L| k_{\parallel}} = \frac{\mathcal{F}_L^2}{4\pi^2 \hbar^4 k_{\parallel}}, \quad (4.5)$$

$$E_{\min} = \hbar\omega_{L_6^-} + \frac{A_u}{\bar{A}_L} (\hbar\omega - \mathcal{E}_g^L), \quad E_{\max} = \hbar\omega_{L_6^-} + \frac{B_u}{\bar{B}_L} (\hbar\omega - \mathcal{E}_g^L), \quad (4.6)$$

with slopes $A_u/\bar{A}_L = 0.745$ and $B_u/\bar{B}_L = 0.896$ for Au. **Here is the clean punchline of the X vs. L comparison:** the $k_{\parallel} \rightarrow 0$ edge — the inverse-square-root singularity of (4.5) — is now the *lower* limit $E_{\min}$, exactly opposite to X where it was the upper limit. Nothing physical was added; the swap is purely because $\mathcal{D}_L < 0$ while $\mathcal{D}_X > 0$. The L line shape is then

$$\mathcal{J}_L(\hbar\omega, T) = \int_{E_{\min}}^{E_{\max}} D_{l \rightarrow u}(E, \hbar\omega) [1 - f(E, T)] dE, \quad (4.7)$$

and its integrable divergence sits right on the onset edge itself. So: the L edge is *sharp and strong*, the X edge *soft and tailed* — this is GRW’s “splitting of the interband absorption edge,” now with both halves coming out of one formula set instead of two separate papers.

Remark 4.1 (A tension in the L level scheme, recorded on purpose) *The fitted levels ($\mathcal{E}_g^L = 2.45$, $\hbar\omega_{L+} = 1.56 \Rightarrow \hbar\omega_{L_6^-} = +0.89$ eV) put the whole window (4.6) above E_F , so the Fermi factor is inert at L and the model's L temperature dependence enters only through broadening and gap shift. But GRW's own qualitative picture of the L singularity — the onset CEDS sitting tangent to the Fermi surface along the neck — needs L_6^- to be below E_F . Both statements cannot hold in one parabolic model. The fitted scheme is the one the code implements; I record the tension here rather than paper over it.*

5 Putting it together: GRW (9)–(10)

Now we cash in. Insert the corrected EDJDOS (3.12) into the FGR result (2.3). The bridge is the definition (3.8) itself, which says $\int d^3k \delta(\Omega - \hbar\omega)[\dots] = (2\pi)^3 \int dE D_{l \rightarrow u}(E, \hbar\omega)[\dots]$ per half-neighborhood; the $(2\pi)^3$ cancels, and

$$\begin{aligned} \epsilon_2(\hbar\omega, T) &= \frac{4\pi^2 e^2 \hbar^2}{3m^2 \omega^2} \cdot 2 \cdot \sum_{i=X,L} N_i |P_i|^2 \mathcal{J}_i(\hbar\omega, T) \\ &= \boxed{\frac{8\pi^2 e^2 \hbar^4}{3m^2 (\hbar\omega)^2} \sum_{i=X,L} N_i |P_i|^2 \mathcal{J}_i(\hbar\omega, T)} \quad [\text{GRW (9)}] \end{aligned} \quad (5.1)$$

— GRW's Eq. (9), the second headline result, recreated with *no leftover constant*. That clean landing is the payoff, and it happens only if three things are all true at once: (i) $|P|^2$ is the squared ∇ matrix element (2.2); (ii) $D_{l \rightarrow u}$ is the *corrected* (3.12), not the printed one; and (iii) spin (the explicit 2) and the half-point counts $N_X = 6$, $N_L = 8$ are kept where shown. (GRW leave the N_i implicit, so their fitted $|P_i|^2$ really means $N_i |P_i|^2$ up to normalization — harmless for the ratio they quote.) The quantities the data actually fix are the strengths,

$$S_X = \mathcal{F}_X |P_X|^2, \quad S_L = \mathcal{F}_L |P_L|^2, \quad [\text{GRW (10)}] \quad (5.2)$$

with $|P_X/P_L|^2 = 0.370$ from their fit to Johnson–Christy. Under the corrected (4') the combination the data really pin down is $N_i \mathcal{F}_i^2 |P_i|^2$; it maps onto GRW's printed S_i through exactly the constant $2\mathcal{F}_i/\hbar^2$ of Remark 5.1, so — to say it one more time — all the published numbers keep their meaning.

Parameters (levels: GRW fit; masses: C&S, with the X row subject to Remark 3.1; $\mathcal{F}$ in units of the free-electron mass):

	$\mathcal{E}_g$ (eV)	upper level (eV)	lower level (eV)	$m_{u\perp}$	$m_{u\parallel}$	$m_{l\perp}$	$m_{l\parallel}$
X (GRW branch)	1.94	$\hbar\omega_{X_6^-} = 0.17$	$\hbar\omega_{X_7^+} = 1.77$	0.19	0.15	0.31	0.40
L	2.45	$\hbar\omega_{L_6^-} = 0.89$	$\hbar\omega_{L+} = 1.56$	0.24	0.12	0.70	1.03

	$\mathcal{D}$ (units $(\hbar^2/2m)^2$)	$\mathcal{F}/m$	window slopes	singular edge
X	+34.7	0.170	$E_{\max}^>: 0.38\text{--}0.62^\dagger$	upper ($k_{\parallel} \rightarrow 0$)
L	−7.86	0.357	$E_{\min}: 0.745, E_{\max}: 0.896$	lower ($k_{\parallel} \rightarrow 0$)

[†] 0.38 with my $\perp$ assignment ($m_{u\perp} = 0.31$), 0.62 with the fully $u \leftrightarrow l$ swapped one ($m_{u\perp} = 0.19$); $\mathcal{F}_X$ is invariant under the full swap. $\mathcal{D}_X$ and $\mathcal{F}_X$ quoted for the full-swap/GRW assignment; my assignment gives the same values by the symmetry of (3.11).

Scorecard — the ten GRW equations, one by one (the three corrected rows are the ones to watch):

GRW	here	status	defect as printed
(1)	(3.1)	recreated	—
(2)	(3.2)	recreated	—
(3)	(3.3)	recreated	—
(4)	(3.12)	corrected	prefactor $(8\pi^2\hbar^2)^{-1}\mathcal{F} \rightarrow (4\pi^2\hbar^4)^{-1}\mathcal{F}^2$; off by $2\mathcal{F}/\hbar^2$ (const.)
(5)	(3.11)	recreated	—
(6)	(3.7)	corrected	dimensionally impossible as printed
(7)	(3.16)	recreated	occupation generalized
(8)	(3.14)	corrected	$u \parallel \leftrightarrow l \parallel$ subscripts swapped
(9)	(5.1)	recreated	N_i made explicit
(10)	(5.2)	recreated	units note (Rem. 5.1)

Remark 5.1 (The constant that hides the (4) discrepancy) *Collecting the (4) discussion in one place, since it is the subtle one. GRW print $D_{l \rightarrow u} = (8\pi^2\hbar^2)^{-1}\mathcal{F}/k_{\parallel}$; the definition (3.8) forces $D_{l \rightarrow u} = (16\pi^2\mathcal{D})^{-1}/k_{\parallel} = \mathcal{F}^2/(4\pi^2\hbar^4 k_{\parallel})$. The ratio of the two is $2\mathcal{F}/\hbar^2$, a pure constant per critical point. It therefore cannot touch any line shape and is absorbed once and for all into the fitted strength $S = \mathcal{F}|P|^2$ of Eq. (5.2). That is exactly why the discrepancy is so easy to miss and why it costs nothing physically — but the FGR chain only closes to (5.1) with unit coefficient if the corrected form is used, which is the independent check in Sec. 6.*

6 Photonic density of states and the emission spectrum

This last section is not part of GRW — it is the bridge to the PL problem I actually care about, and it doubles as a global check on every prefactor above.

6.1 Mode counting

Two transverse polarizations, $\omega = cq$, periodic quantization:

$$\rho(\omega) = \frac{2}{(2\pi)^3} 4\pi q^2 \frac{dq}{d\omega} = \boxed{\frac{\omega^2}{\pi^2 c^3}} \quad (6.1)$$

modes per unit volume per unit angular frequency (in a host of index n , $c \rightarrow c/n$; in a structured environment $\rho \rightarrow$ the local density of states $\rho(\mathbf{r}, \omega)$ — which is the photonic factor of the PL factorization).

6.2 Spontaneous emission of one pair

Quantize $\mathbf{A}$ (mode amplitudes $\sqrt{2\pi\hbar c^2/\omega V}$ in Gaussian units) and apply FGR at photon vacuum, using $\sum_{\lambda} \int d\Omega_q |\hat{\mathbf{e}}_{\lambda} \cdot \mathbf{p}_{ul}|^2 = \frac{8\pi}{3} |\mathbf{p}_{ul}|^2$ and $|\mathbf{p}_{ul}|^2 = \hbar^2 |P|^2$:

$$W_{\text{sp}}(\omega) = \frac{4e^2\hbar\omega |P|^2}{3m^2c^3} = \underbrace{\frac{4\pi^2 e^2\hbar |P|^2}{3m^2\omega}}_{\text{matter}} \times \underbrace{\frac{\omega^2}{\pi^2 c^3}}_{\rho(\omega)}, \quad (6.2)$$

the Einstein A coefficient in the $\mathbf{A} \cdot \mathbf{p}$ form, with the photonic density of states sitting there as an exact factor. (Note the semiclassical field of Sec. 2 can never produce (6.2): it is the $+1$ in $\langle n+1 \rangle$ — a genuine vacuum effect, which is why spontaneous emission needs the quantized field and cannot fall out of the classical calculation.)

6.3 The interband PL spectrum, and the closure check

Sum over emitting pairs of one critical-point family (spin 2, N half-neighborhoods, the same EDJDOS (3.8)):

$$R(\hbar\omega) = \rho(\omega) \frac{4\pi^2 e^2 \hbar}{3m^2 \omega} 2N |P|^2 \int dE f(E, T) [1 - f(E - \hbar\omega, T)] D_{l \rightarrow u}(E, \hbar\omega) \quad (6.3)$$

photons per unit time, volume and photon energy — *same* $D_{l \rightarrow u}$, same windows and edges as absorption; only the occupation factor is different. Now use the equilibrium identity $f(E)[1 - f(E - \hbar\omega)] = n_B(\hbar\omega) [f(E - \hbar\omega) - f(E)]$: every electronic factor maps back onto (5.1), the constants cancel *exactly*, and

$$R(\hbar\omega) = \frac{\omega^3}{\pi^2 \hbar c^3} \epsilon_2(\omega) n_B(\hbar\omega) = \frac{\omega \rho(\omega)}{\hbar} \epsilon_2(\omega) n_B(\hbar\omega) \quad (6.4)$$

— the van Roosbroeck–Shockley relation, i.e. Kirchhoff’s law in the per-mode form $R_{\text{em}}/R_{\text{abs}} = e^{-\hbar\omega/k_B T}$. **This is the global check I promised.** Equations (5.1) and (6.3) were assembled independently, yet they satisfy (6.4) with coefficient exactly one — which pins down every prefactor in the document at a stroke. And it would fail, by exactly that constant $2\mathcal{F}/\hbar^2$, if the printed (4) were used with any fixed matrix-element convention: so the (4) correction is not a matter of taste, it is what makes this identity close. Out of equilibrium ($f \rightarrow f^S, f^P$) the identity breaks, and the *deviation* from (6.4) is precisely the non-equilibrium PL signal I am after; in a nanostructure, $\rho(\omega) \rightarrow \rho(\mathbf{r}, \omega)$ in (6.3).

A Dimensional audit (Gaussian)

Keeping myself honest. $[e^2] = \text{erg cm}$, $[|P|^2] = \text{cm}^{-2}$, $[\mathcal{D}] = \text{erg}^2 \text{cm}^4$, $[\mathcal{F}] = \text{g}$, $[D_{l \rightarrow u}] = \text{erg}^{-2} \text{cm}^{-3}$ (states per volume per energy²), $[\mathcal{J}] = \text{erg}^{-1} \text{cm}^{-3}$.

- (3.12): $\mathcal{F}^2/\hbar^4 k_{\parallel} \sim \text{g}^2/(\text{erg}^4 \text{s}^4 \text{cm}^{-1}) = \text{erg}^{-2} \text{cm}^{-3}$. ✓
- (5.1): $\frac{\text{erg cm} \cdot \text{erg}^4 \text{s}^4}{\text{g}^2 \text{erg}^2} \cdot \text{cm}^{-2} \cdot \text{erg}^{-1} \text{cm}^{-3} = \text{erg}^2 \text{s}^4 \text{g}^{-2} \text{cm}^{-4} = 1$. ✓
- (6.2): $\frac{\text{erg cm} \cdot \text{erg s} \cdot \text{s}^{-1} \cdot \text{cm}^{-2}}{\text{g}^2 \text{cm}^3 \text{s}^{-3}} = \text{s}^{-1}$. ✓
- (6.3): $[\rho][\cdot] \sim \text{s cm}^{-3} \cdot \text{s}^{-1} \cdot \text{erg}^{-1} \cdot \dots = \text{s}^{-1} \text{erg}^{-1} \text{cm}^{-3}$: photons/(s erg cm³). ✓
- (6.4): $[\omega \rho/\hbar] = \text{s}^{-1} \cdot \text{s cm}^{-3} \cdot \text{erg}^{-1} \text{s}^{-1} = \text{erg}^{-1} \text{cm}^{-3} \text{s}^{-1}$. ✓

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